

consumable biomedical test equipment, and so on.

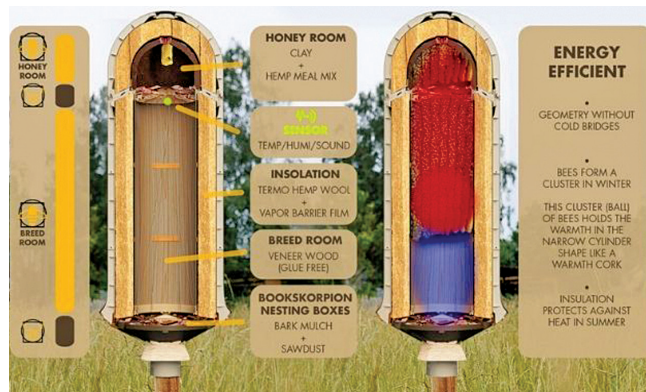
3D printing plus Arduino is an all-time favourite combo of hobbyists, and you can find several robots ranging from cute to chunky on Instructables and Thingiverse. Do check out the lovable dancing robot Tito, the spider-like Hexapoduino, the handy MeArm V0.4, and more such projects online.

Others. You eat the food with your eyes first, goes the adage. It is therefore not surprising that the food industry has been quick to pick up 3D printing techniques to create visually-appealing foods like never before. 3D printing is used widely to produce small batches of customised foods, and for creating personalised nutrition packages.

3D printing is also a wonderful learning tool. Around the world, schools and colleges are using 3D printing to print a range of models from historic monuments to functional parts, to help students understand concepts better.

It is also finding innovative uses in farms. In Germany, for example, a startup called HIIVE is 3D printing tree cavities for bee farms. Honeybees like the *Apis Mellifera* originally live in tree cavities, which have a special microclimate that is not only good for bees but also for beneficial symbionts that co-inhabit the tree cavities with the bees.

Using a close-to-natural hive, bee farms can attract more bees, and get a better harvest, while also ensuring a natural habitat for the bees. A symbiotic deal indeed. 3D printing made it possible for HIIVE to iteratively tweak



German company HIIVE 3D-prints tree cavities to provide a natural home for bees in farms (Courtesy: HIIVE)

the design to develop an optimal structure for the tree cavities, which are 3D-printed with multiple natural materials.

3D printing also enables a lot of innovation and optimisation in the electronics industry. Rapid prototyping and the ability to manufacture small batches of electronics at a lower cost is a boon for all consuming industries. 3D printing is also said to be more eco-friendly, as it generates much less e-waste as compared to the traditional etching processes. The ability to print on non-flat surfaces and on varied materials, including plastics, opens up new opportunities for innovation.

Space Foundry, a company launched by former NASA researcher Ram Prasad Gandhiraman and Dennis Nordlund of Stanford University, has created a plasma based 3D printing process, which can print electronics in a single-step approach that does not require heat or ultraviolet curing. The multilateral printing platform can print various materials, including metals like copper, dielectrics, organics and even bioinks. It is an environment-friendly process that does not leave behind liquid industrial waste or toxic by-products.

NASA believes this tech can enhance its in-space manufacturing efforts through on-demand manufacturing of electronics (ODME), as plasma jet printing can potentially be used for manufacturing in the low Earth orbit

and on the lunar surface. In a press report, Gandhiraman describes their 3D printer as a highly miniaturised version of semiconductor manufacturing technology.

As a service. While we do see random use of 3D printing in almost all industries, it is yet to gain traction in mission-critical applications, as certification and part qualification processes are still not in

place. For rapid prototyping and testing, or for one-off production needs, it does not make sense for small companies to invest in 3D printing setups. However, everyone would be happy to have a go at it, if available as a service. There are many companies offering 3D printing as a service, and this is likely to remain the popular choice in the near future too.

3D printing just right for Atmanirbhar Bharat

Recognising the scope of 3D printing, the Ministry of Electronics and Information Technology (MeitY) released the National Strategy for Additive Manufacturing earlier this year. The ministry hopes to increase India's share in the global additive manufacturing to 5% by 2025. The national strategy sets various goals for the industry in this direction—to add US\$1 billion to the gross domestic product; develop 50 India-specific technologies for material, machine and software; setup 100 new start-ups for additive manufacturing; develop 500 new products; train at least 100,000 new skilled workers, and so on, in the next three years.

The strategy recommends that India must come forth and adopt additive manufacturing in all sectors, including defence and public sectors. Adoption of 3D printing on an industrial scale could help domestic companies to overcome technical and economic barriers—and tread proudly on the path of a self-reliant India! **EFY**

3D Printer in Space

According to a recent news report, aerospace company Airbus is planning to send a metal 3D printer, Metal3D, to the International Space Station (ISS) in 2023, as part of its plans to establish an orbital satellite factory. According to Airbus, this will be the first metal 3D printer on the ISS. It will enable astronauts to print tools and parts like radiation shields. The company also says in the media report that future versions of the printer will be able to create objects out of lunar soil and recycle parts from decommissioned satellites onboard an orbital satellite factory.